Available online at www.sciencedirect.com
ScienceDirect
Electrochemistry Communications 9 (2007) 703–707
electrochemistry communications
www.elsevier.com/locate/elecom

Bifunctional electrolyte additive for lithium-ion batteries ☆

Zonghai Chen, K. Amine *

Argonne National Laboratory, Electrochemical Technology Program, 9700 South Cass Avenue, Argonne, IL 60439, USA

Received 24 October 2006; received in revised form 1 November 2006; accepted 2 November 2006
Available online 4 December 2006

Abstract

2-(Pentafluorophenyl)-tetrafluoro-1,3,2-benzodioxaborole was reported as a bifunctional electrolyte additive for lithium-ion batteries. It was found that the reported additive had a redox potential of 4.43 V vs. Li^+/Li with a reversible oxidation/reduction reaction. Therefore, it is a promising redox shuttle for overcharge protection of most positive electrode materials for current lithium-ion-battery technology. At the same time, the boron center of this additive is a strong Lewis acid and can act as an anion receptor to dissolve LiF generated during the operation of lithium-ion batteries. The possibility of using the novel additive as both the redox shuttle and the anion receptor was discussed.

Keywords: Redox shuttle; Anion receptor; Electrolyte additive

1. Introduction

Overcharge is one type of abuse to lithium-ion cells by forcing a current through to charge the cell when the lithium-ion cell is already fully charged. Overcharging a lithium-ion cell can generally result in the failure of a battery pack or a fire hazard. In order to maintain the normal operation of a battery pack, overcharge protection mechanisms need to be incorporated into each cell to prevent the unexpected overcharge of one or more lithium-ion cells. A redox shuttle is one of the proposed protection mechanisms for lithium-ion batteries [1–8]. The redox shuttle is an electrolyte additive that can be reversibly oxidized/reduced at a characteristic potential and provides an intrinsic overcharge protection. Once extra charge is forced through a lithium-ion cell containing the redox shuttle, the redox shuttle is oxidized on the positive electrode; the oxidized specie diffuses back to the negative electrode and is reduced to its original state. Under this mechanism, the extra charge can be shuttled through the lithium-ion cells by the redox reaction of the redox shuttle without causing any damage to the lithium-ion cells.

The redox shuttles reported in the literature include ferrocene [1] and aromatic compounds [2–5]. Due to the low redox potential of ferrocene (generally less than 3.5 V vs. Li^+/Li), it is not practical to incorporate it into state-of-the-art lithium-ion batteries. The aromatic compounds, whose redox potential can be as high as 4.2 V vs. Li^+/Li, were first patented by Sony corporation [2,3]. Lately, it was reported that the compounds proposed by Sony corporation were not stable at all in the lithium-ion-battery environment [4]. Alternatively, Chen et al. [4] reported a stable redox shuttle, 2,5-ditertbutyle-1,4-dimethoxybenzene, that can sustain more than 200 cycles with 100% overcharge for each cycle. However, 2,5-ditertbutyl-1,4-dimethoxybenzene has a relatively low redox potential (~3.9 V vs. Li^+/Li), and is only practical for overcharge protection of $LiFePO_4$. A desired redox shuttle is expected to have a

☆ The submitted manuscript has been created by UChicago Argonne, LLC, Operator of Argonne National Laboratory (“Argonne”). Argonne, a US Department of Energy Office of Science laboratory, is operated under Contract No. DE-AC02-06CH11357. The US Government retains for itself, and others acting on its behalf, a paid-up nonexclusive, irrevocable worldwide license in said article to reproduce, prepare derivative works, distribute copies to the public, and perform publicly and display publicly, by or on behalf of the Government.

* Corresponding author. Fax: +1 630 252 4176.
E-mail address: amine@cmt.anl.gov (K. Amine).

1388-2481/$ - see front matter © 2006 Elsevier B.V. All rights reserved.
doi:10.1016/j.elecom.2006.11.002

redox potential between 4.3 V and 4.5 V vs. Li^+/Li, to be compatible with the working potential of major positive electrode materials like $LiCoO_2$ and $Li[Ni_xMn_yCo_{1-x-y}]O_2$ ($0 \leqslant x, y, x+y \leqslant 1$). The upper limit potential of 4.5 V is desired to ensure that the overcharge protection mechanism be activated before the decomposition of the electrolyte components, such as carbonates that can be oxidized at 4.8 V vs. Li^+/Li.

In this work, 2-(pentafluorophenyl)-tetrafluoro-1,3,2-benzodioxaborole was reported as a stable redox shuttle with a redox potential of 4.43 V vs. Li^+/Li. Moreover, the novel additive is a strong Lewis acid, and can act as an anion receptor as well. The impact of an anion receptor on the capacity retention and the power capability of lithium-ion cells has been reported previously [9]. In this paper, we discuss the potential of using 2-(pentafluorophenyl)-tetrafluoro-1,3,2-benzodioxaborole as a bifunctional additive for overcharge protection and improving the life of lithium-ion cells.

2. Experimental

The bifunctional additive investigated is 2-(pentafluorophenyl)-tetrafluoro-1,3,2-benzodioxaborole (PFPTFBB), whose molecular structure is shown in Fig. 1. The synthesis route for PFPTFBB is summarized in what follows. 0.91 g 3,4,5,6-tetrafluoro-1,2-dihydroxybenzene and 1.05 g 2,3,4,5,6-pentafluorobenzenbornic acid were mixed in 10 mL anhydrous toluene. The mixture was then heated and refluxed for more than 5 h. The water generated during the condensation reaction was removed by azeotropic distillation. After reflux, the solvent was evaporated from the reaction mixture under reduced pressure at a low temperature (<80 °C). Pure-compound, needle-shape colorless crystals were obtained by subliming the residues at 110 °C and reduced pressure (~0.1 mm Hg).

The basic electrochemical properties of PFPTFBB were characterized with cyclic voltammetry using an IM6 electrochemical impedance spectrometer and a Pt/Li/Li three-electrode cell. The overcharge protection performance of this redox shuttle was tested in coin cells. The cell stack consisted of a negative electrode, a microporous polypropylene separator (Celgard 3501), a positive electrode, and an appropriate amount of electrolyte. The cells were charged using a constant current to 200% of their normal capacity (100% overcharge) or until a specific upper cutoff voltage (i.e., 4.95 V) was reached, whichever occurred first. After the charging process, the cells were discharged to 3.0 V with the same constant current.

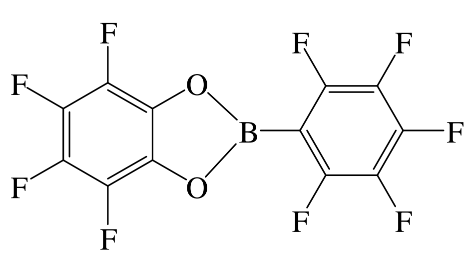

Fig. 1. Molecular structure of 2-(pentafluorophenyl)-tetrafluoro-1,3,2-benzodioxaborole.

3. Results and discussion

Fig. 2 shows cyclic voltammograms of 0.05 M PFPTFBB and 1.2 M $LiPF_6$ in ethylene carbonate (EC)/ethyl methyl carbonate (EMC) (3:7, by weight). Both the counter and reference electrodes were lithium foils, and the working electrode was a platinum disk ($\Phi = 1$ mm). Fig. 2 shows that PFPTFBB has a reversible redox reaction at about 4.43 V vs. Li^+/Li. The onset potential of PFPTFBB is about 4.3 V vs. Li^+/Li, which is high enough to provide overcharge protection for most state-of-the-art positive electrode materials. It was also found that the anionic peak current (I_p) increased linearly with the square root of the scanning rate ($v^{1/2}$). Eq. (1) gives a theoretical description on the relationship between I_p and $v^{1/2}$.

$$I_p = 2.69 \times 10^5 n^{3/2} A D^{1/2} v^{1/2} C \qquad (1)$$

where A is the surface area of the working electrode (cm^2), n is the number of electrons involved in the redox process ($n = 1$ for this case), C is the concentration of active material in the solution (mol/cm^3), and D is the diffusion coefficient (cm^2/s). Based on Eq. (1), the diffusion coefficient is estimated to be 1.4×10^{-6} cm^2/s.

Fig. 3 shows the voltage profile of two identical graphite/$LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$ lithium-ion cells during the initial overcharge testing cycle. The electrolyte used was 1.2 M $LiPF_6$ in EC/PC/DMC (1:1:3, by weight) with 5 wt% PFPTFBB. At the beginning, the cells were normally charged with a constant current of C/10. Both cells were fully charged in about 10 h, and the voltage of both cells continued to increase until the positive potential met the redox potential of PFPTFBB, at which point the cell

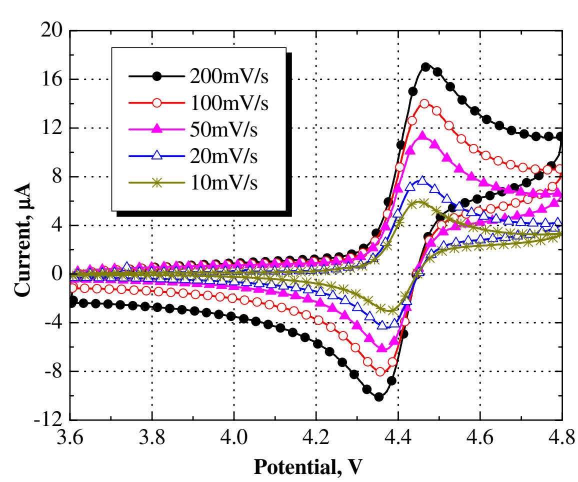

Fig. 2. Cyclic voltammograms of 0.05 M PFPTFBB and 1.2 M $LiPF_6$ in EC/EMC (3:7, by weight) using a Pt/Li/Li three-electrode cell.

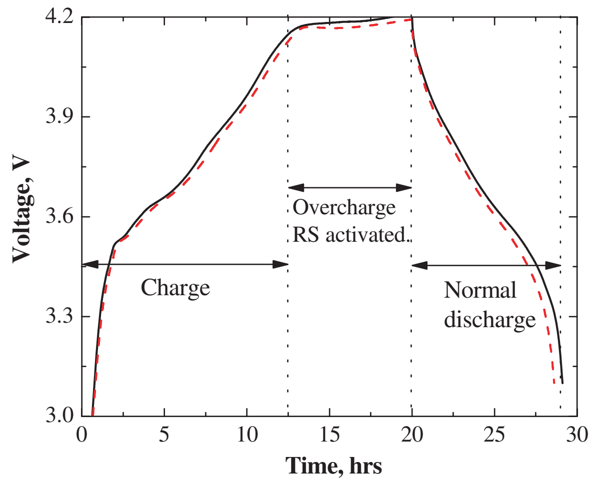

Fig. 3. Voltage profiles of two graphite/$LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$ lithium-ion cells during the overcharge test.

voltage was about 4.2 V. Fig. 3 shows a flat voltage at 4.2 V for 10-h overcharge, because the redox reaction of PFPTFBB was activated at this point, and the shuttle mechanism of PFPTFBB continuously shuttled the charge, which was forced by the charger through the cell, without causing any cell damage or intercalation/deintercalation reaction.

Fig. 4 shows the charge/discharge capacity of a graphite/$LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$ lithium-ion cell containing 5 wt% PFPTFBB in the electrolyte, which was 1.2 M $LiPF_6$ in EC/PC/DMC (1:1:3, by weight). The cell was charged and discharged at a constant current of C/10 (0.2 mA). During the charging process, the cell was charged to 4.95 V or until 4.0 mAh charge was delivered (100% overcharge). The cell was initially tested at 25 °C for 20 cycles and was then heated in an oven to 55 °C for another 50 cycles to check the stability of redox shuttle under a more aggressive testing condition. After that, the cell was tested with a higher constant current of C/5. The testing condition for each stage was also labeled in Fig. 4. During the initial cycling at 25 °C, some fluctuation of the discharge capacity was observed. The voltage profiles of the cell were checked, and no evidence of over-discharge was observed. The unusually high discharge capacity, up to 3.0 mAh during the initial cycling is not yet fully understood. After 50 cycles at 55 °C, the overcharge protection of the redox shuttle was maintained and the cell capacity remained very stable, even though the cell was 100% overcharged for each cycle. Afterward, the cell was further tested at 25 °C and 55 °C, respectively with a constant current of C/5 and 100% overcharge. The overcharge protection provided by PFPTFBB finally disappeared after 170 cycles of 100% overcharge. Fig. 4 also shows the cell completely lost its capacity after 125 cycles. However, the redox shuttle mechanism remained active for another 50 cycles at 55 °C after the cell died. Therefore, PFPTFBB is believed to be a stable redox shuttle for overcharge protection of 4-V class lithium-ion batteries.

It has been reported that the electrochemical reactivity of PFPTFBB is provided by the phenyl group directly attached to the oxygen atoms, while the fluorine substitution is adopted to raise its redox potential and to kick off the potential polymerization upon oxidation [5]. The second sub group of PFPTFBB (pentafluorophenyl borole) has no electrochemical reactivity, and its structure is borrowed from tris(pentafluoro phenyl)borane (TPFPB), which has been previously reported as an anion receptor to improve the capacity retention and power capability of lithium-ion batteries [9]. Therefore, PFPTFBB can act as an anion receptor as well as a redox shuttle. Fig. 5 shows the discharge capacity of MCMB/$Li_{1.1}[Mn_{1/3}Ni_{1/3}Co_{1/3}]_{0.9}O_2$ lithium-ion cells that were cycled at 55 °C with a constant current of C/2 (~1.0 mA). The initial specific capacity of the cells is about 128 mAh/g. The electrolytes used were 1.2 M $LiPF_6$ in EC/EMC (3:7, by weight) without and with

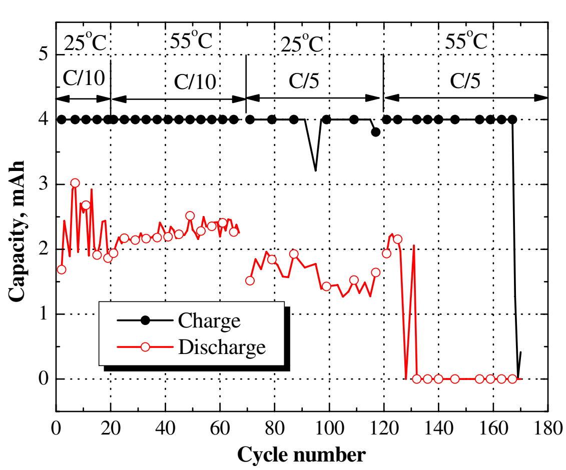

Fig. 4. Charge and discharge capacity of a graphite/$LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$ lithium-ion cell during the whole course of overcharge test. The electrolyte used contained 5 wt% PFPTFBB.

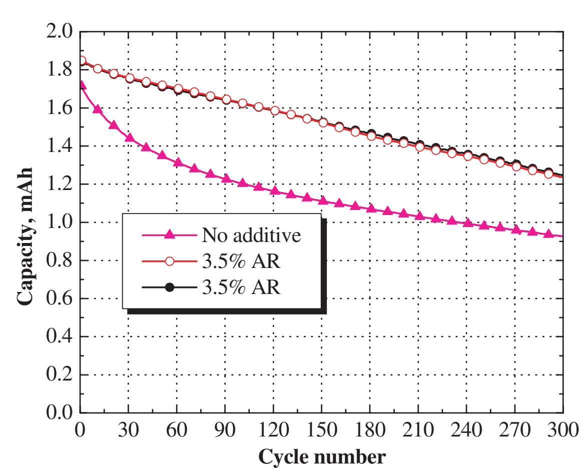

Fig. 5. Capacity retention of MCMB/$Li_{1.1}[Ni_{1/3}Co_{1/3}Mn_{1/3}]_{0.9}O_2$ cells showing the positive impact of PFPTFBB as an anion receptor. The cells were cycled between 3.0 V and 4.0 V with a constant current of C/2 at 55 °C.

3.5 wt% PFPTFBB. Two cells with 3.5 wt% PFPTFBB were tested; their performance was identical to each other as shown in Fig. 5. This Fig. 5 shows that the addition of PFPTFBB greatly improved the capacity retention of lithium-ion cells. The improvement on the cell capacity retention was attributed to dissolving inorganic components, such as LiF, from the passivation films by the anion receptors, such as PFPTFBB and TPFPB (see Eq. (2)).

$$\mathrm{PFPTFBB} + \mathrm{LiF} \rightarrow \mathrm{Li}^+ + \mathrm{PFPTFBB} - \mathrm{F}^- \tag{2}$$

Generally, the amount of LiF presenting in the passivation films is very small, and the amount of the anion receptor needed to dissolve the LiF is also small. When excess amount of anion receptor, i.e., PFPTFBB, was added to the electrolyte, it would accelerate the decomposition of $LiPF_6$, as in Eq. (3).

$$\mathrm{PFPTFBB} + \mathrm{PF}_6^- \rightleftharpoons \mathrm{PFPTFBB} - \mathrm{F}^- + \mathrm{PF}_5 \tag{3}$$

When the reaction reaches its thermodynamic equilibrium, the concentration of PF_5 ($[PF_5]$) can be expressed as the following formula (4).

$$[\mathrm{PF}_5] = \frac{k[\mathrm{PFPTFBB}][\mathrm{PF}_6^-]}{[\mathrm{PFPTFBB} - \mathrm{F}^-]} \tag{4}$$

In above formula, $[X]$ is the concentration of the specie X, and k is a thermodynamic constant. Apparently, the concentration of PF_5 increases linearly with the excess amount of PFPTFBB added, while PF_5 was well known to initiate a ring opening polymerization of carbonates [10], resulting in performance degradation. For instance, we have reported that excess amount of TPFPB can significantly increase the interfacial impedance of lithium-ion cells [9].

Based on the above arguments, only a controlled amount of PFPTFBB is needed to act as an anion receptor to improve the power capability of the lithium-ion cells. However, a high-concentration of the redox shuttle is desired to maximize its overcharge protection capability. Therefore, this conflict needs to be addressed before one can integrate the functionality of the anion receptor and the redox shuttle into a single molecule.

Eq. (4) suggests that an alternative practical parameter to control the concentration of PF_5 is PFPTFBB–F^-, which is believed to have the same redox shuttle capability as PFPTFBB. In order to confirm this assumption, another cyclic voltammogram (CV) was carried out by adding equal mole of LiF to the electrolyte with 0.05 M PFPTFBB, assuming that all PFPTFBB was converted to PFPTFBB–F^-. Fig. 6 shows the CV of the electrolyte with and without LiF addition. Apparently, the addition of LiF, or converting PFPTFBB to PFPTFBB–F^-, doesn't impact the electrochemical performance of the additive in terms of the redox potential and the shuttle current density it can provide. Therefore, the concentration of PF_5 in the electrolyte can be controlled by varying the ratio of PFPTFBB to LiF added to the electrolyte. In this case, the maximum shuttle current that can be provided by the additive is determined by the amount of PFPTFBB added regardless to the amount of LiF added. Meanwhile, the excess PFPTFBB not converting to PFPTFBB–F^- acts as an anion receptor for dissolving LiF. An example to integrate both functionalities is a combination of high-concentration PFPTFBB–F^- (3–5 wt%) and low-concentration PFPTFBB ($\leqslant$ 1 wt%). In this example, PFPTFBB–F^- can be obtained by adding equal mole of PFPTFBB and LiF; both PFPTFBB–F^- and PFPTFBB are redox shuttles, and only PFPTFBB is the active anion receptor.

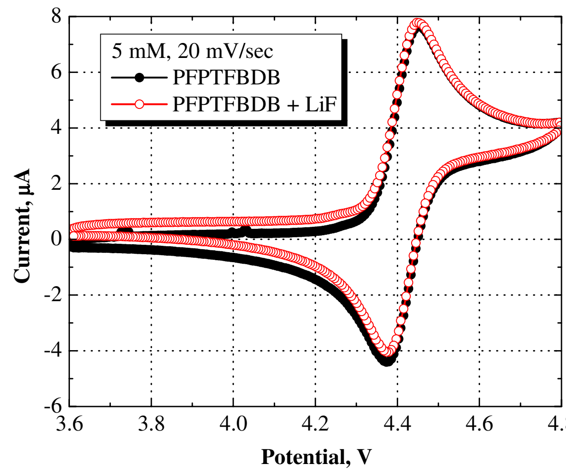

Fig. 6. Cyclic voltammograms of 0.05 M PFPTFBB and 1.2 M $LiPF_6$ in EC/EMC (3:7, by weight) showing the electrochemical reactivity of PFPTFBB–F^-.

4. Conclusion

2-(Pentafluorophenyl)-tetrafluoro-1,3,2-benzodioxaborole was investigated as both a redox shuttle and an anion receptor for lithium-ion batteries. As redox shuttles, both PFPTFBB and PFPTFBB–F^- show a reversible redox reaction at 4.43 V vs. Li^+/Li, which is high enough for overcharge protection of state-of-the-art positive electrode materials. The redox potential of 4.4 V is very suitable for repeating cell overcharging without degrading the electrolyte system. This can also ease the cell balancing in a battery pack. PFPTFBB can also be used as an anion receptor to improve the capacity retention of lithium-ion cells by dissolving the deposited LiF in the passivation films. It is also suggested that the power capability of the lithium-ion cells can be improved by using the combination of PFPTFBB and LiF. Finally, further investigation of the electrochemical stability of PFPTFBB–F^- anion as a redox shuttle is ongoing.

Acknowledgement

The authors acknowledge the financial support of the US Department of Energy, Freedom CAR and Vehicle Technologies Program, under Contract No. DE-AC02-06CH11357.

References

[1] K.M. Abraham, D.M. Pasquariello, E.B. Willstaedt, J. Electrochem. Soc. 137 (1990) 1856–1857.
[2] M. Adachi, US Pat. No. 5,763,119 (1998).
[3] M. Adachi, K. Tanaka, K. Sekai, J. Electrochem. Soc. 146 (1999) 1256–1261.
[4] J. Chen, C. Buhrmester, J.R. Dahn, Electrochem. Solid-State Lett. 8 (2005) A59–A62.
[5] Z.H. Chen, Q.Z. Wang, K. Amine, J. Electrochem. Soc. 153 (2006) A2215–A2219.
[6] J.R. Dahn, J. Chen, C. Buhrmester, U.S. Pat. Pub. No. 2005/0221196 A1 (2005).
[7] J.R. Dahn, J. Chen, U.S. Pat. Pub. No. 2005/0221168 A1 (2005).
[8] R.L. Wang, J.R. Dahn, J. Electrochem. Soc. 153 (2006) A1922–A1928.
[9] Z. Chen, K. Amine, J. Electrochem. Soc. 153 (2006) A1221–A1225.
[10] S.E. Sloop, J.K. Pugh, S. Wang, J.B. Kerr, K. Kinoshita, Electrochem. Solid-State Lett. 4 (2001) A42–A44.